

ECON 230 — Introduction to Behavioural Economics

Course Textbook

Chapter 10

From Lab to Policy — Applications and Behavioural Policy Design

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Chapter Overview

This final chapter of the textbook asks the most practically important question in behavioural economics: what should we *do* with what we have learned?

The preceding nine chapters have established a rich picture of human decision-making that departs systematically from the rational agent benchmark. People use heuristics that introduce predictable biases. They are loss averse, reference-dependent, and probability-distorted in ways captured by prospect theory. They discount the future hyperbolically, plan to save and exercise and eat well, and then fail to follow through because of present bias. They are powerfully influenced by the choice architecture — the defaults, framing, and social norms — that surrounds every decision. They have social preferences that make them willing to incur personal costs to act fairly, to punish norm violators, and to contribute to collective goods. In financial markets, these individual-level tendencies aggregate into anomalies, excess volatility, momentum, and periodic bubbles that cannot be reconciled with the efficient market hypothesis in its strongest form.

These findings are scientifically important. But their greatest value lies in what they imply for institutions, policy, and practice. If people systematically undersave for retirement, what should pension regulation look like? If present bias causes people to smoke cigarettes they later wish they had not smoked, should policy intervene — and if so, how? If tax compliance responds strongly to social norms, what should tax letters look like? If the decision environment in financial markets systematically disadvantages retail investors, what does good financial regulation require?

This chapter develops a framework for answering these questions. It begins with the methodology of evidence-based policy — particularly the randomised controlled trial — as a tool for translating laboratory findings into field interventions. It then examines six major policy domains in depth: tax compliance, health behaviour, consumer finance, environmental behaviour, international development, and education. In each domain, it applies the behavioural insights developed in earlier chapters to real policy problems, reviews the evidence on what works, and identifies the institutional and design questions that remain open.

The chapter closes with three sets of cross-cutting themes: the comparative analysis of nudges, incentives, and mandates as policy instruments; the limits of behavioural economics and the conditions under which it is insufficient; and the open questions at the frontier of

the field that will occupy researchers and policymakers over the coming decades.

Learning Objectives

After studying this chapter, students should be able to:

1. **Describe** the randomised controlled trial (RCT) as a method for evaluating behavioural interventions and explain why it is the preferred method for causal identification.
2. **Apply** the behavioural policy cycle — from lab finding to field pilot to scaled implementation — to a specific policy problem.
3. **Explain** how social norm messaging has been applied to tax compliance, citing the Hallsworth et al. (2017) evidence and quantifying its effects.
4. **Identify** at least four behavioural mechanisms that impede healthy behaviour and describe specific interventions targeting each.
5. **Explain** the limits of disclosure regulation as a consumer protection tool and describe behavioural alternatives.
6. **Describe** the role of defaults, carbon labels, and social norms in environmental policy and evaluate the evidence on their effectiveness.
7. **Explain** the application of commitment savings and conditional cash transfers to development economics, citing specific field experiments.
8. **Distinguish** between hard paternalism, soft paternalism, libertarian paternalism, and standard economics in terms of their assumptions, policy recommendations, and ethical commitments.
9. **Identify** at least three conditions under which nudges are insufficient or counterproductive and explain what stronger interventions are needed.
10. **Evaluate** the major open questions in behavioural economics — replication, cultural generalisability, long-run effects, and the role of institutions — and articulate an informed view on the future of the field.

Introduction: The Distance Between Lab and Life

In 1979, Daniel Kahneman and Amos Tversky published their landmark paper introducing Prospect Theory. In that paper, they presented experimental subjects in a university psychology laboratory with hypothetical choices between gambles. The subjects made these choices by marking responses on a paper form. The choices had no financial consequences — they were purely hypothetical, designed to reveal the structure of preferences.

From these pencil-and-paper exercises, Kahneman and Tversky derived findings that would eventually reshape how economists think about financial markets, legal judgements, medical decisions, insurance, retirement savings, and public policy. That distance — from a laboratory experiment with hypothetical stakes to the redesign of pension systems affecting tens of millions of people — is the central methodological challenge of applied behavioural economics.

How confident can we be that laboratory findings predict field behaviour? Laboratory subjects are often university students; real policy targets are diverse populations. Laboratory choices are typically hypothetical; real decisions have stakes. Laboratory designs are controlled and clean; real contexts are messy and confounded. Does a finding about preference reversals in a psychology laboratory translate into a prediction about how workers will respond to pension defaults?

The answer, developed over three decades of field research, is: *sometimes yes, sometimes no, and the only way to know for certain is to run the experiment in the field.*

This insight — that the empirical translation of behavioural findings requires rigorous field testing — is perhaps the most important methodological contribution of applied behavioural economics. The discipline does not simply export laboratory findings to policy. It develops tools — particularly the **randomised controlled trial (RCT)** — for testing whether laboratory-derived interventions work in real policy contexts, with real stakes, in diverse populations.

This chapter examines what we have learned from that process of translation: which laboratory insights have proved robust and generalisable, which have failed to replicate or scale, and what this tells us about both the scope and the limits of the behavioural approach to policy.

1 1. The Empirical Policy Cycle: From Experiment to Implementation

1.1 1.1 The Randomised Controlled Trial in Policy

The **randomised controlled trial** (RCT) is the methodological gold standard for causal inference in policy evaluation. Its logic is simple: to determine whether intervention X causes outcome Y, randomly assign units (individuals, households, schools, businesses) to a treatment group (which receives X) and a control group (which does not), and compare outcomes.

Random assignment ensures that, on average, treatment and control groups are identical in all characteristics — observed and unobserved — except for the receipt of treatment. Any systematic difference in outcomes between the groups can therefore be attributed to the treatment rather than to pre-existing differences between groups.

Before RCTs became standard, policy evaluation was dominated by **observational studies**: researchers observed naturally occurring differences in exposure to a policy and tried to control for confounding factors statistically. The problem is that exposure to a policy is rarely random — it is typically correlated with many other features of the individual or environment. Workers who choose to enrol in a pension plan differ in many ways from workers who do not; comparing their savings outcomes confounds the effect of the plan with the effect of the underlying differences.

The rise of RCTs in economics — often called the "credibility revolution" and associated with the 2019 Nobel Prize awarded to Abhijit Banerjee, Esther Duflo, and Michael Kremer — has transformed policy evaluation. Applied behavioural economics has been at the forefront of this movement, both because its laboratory roots make it natural to think in experimental terms and because its policy focus requires demonstrating that interventions work in the real world.

1.2 1.2 The Behavioural Policy Cycle

A mature behavioural policy programme moves through several stages:

Stage 1 — Laboratory finding. A psychological regularity is identified in controlled experiments. Example: people exhibit status quo bias (they prefer the current state to alternatives even when switching would benefit them).

Stage 2 — Theoretical mechanism. An economic model explains *why* the regularity occurs and *when* it is most likely to matter. Example: status quo bias reflects loss aversion (deviations from the reference point feel like losses) and present bias (switching requires cognitive effort that is costly now).

Stage 3 — Field pilot. The intervention is tested in a real-world context with real stakes, using a randomised design. Example: Madrian and Shea (2001) observe that changing the pension default from opt-in to automatic enrolment raises participation from 49% to 86%.

Stage 4 — Replication and variation. The field pilot is repeated in different contexts, with different populations, to test generalisability. Example: automatic enrolment effects are replicated across many firms, industries, and countries.

Stage 5 — Scale-up and evaluation. The intervention is implemented at policy scale, with ongoing monitoring of effects and unintended consequences. Example: the US Pension Protection Act (2006) encouraged automatic enrolment across the private sector; the UK auto-enrolment mandate (2012) extended it to all employers.

Stage 6 — Iteration. Findings from scale-up feed back into theoretical models and design iterations. Example: the persistence of low default contribution rates across many employees triggers the development of SMarT and other escalation programmes.

This cycle is iterative, not linear: findings at Stage 5 frequently send researchers back to Stage 2 to refine theoretical models.

1.3 1.3 Common Pitfalls in Scaling Behavioural Interventions

The history of applied behavioural economics includes many cases where promising laboratory findings failed to replicate at scale or produced unintended consequences. Several patterns recur:

The Hawthorne effect. When people know they are being observed or studied, they change their behaviour — sometimes in the direction of the treatment, sometimes not. Laboratory and pilot study participants may respond differently from people in scaled-up programmes where they are unaware of being in an experiment.

Publication bias. Academic journals publish positive findings more readily than null results. This means the published literature overrepresents successful interventions. A meta-analysis that draws only on published studies will overestimate the average effect of behavioural interventions.

Context dependence. An intervention that works in one cultural, institutional, or temporal context may not work in another. The social norm nudge that increased tax compliance in the UK required modifications for effective application in Guatemala. Cafeteria designs that increased healthy eating in a US university cafeteria did not replicate in an Austrian secondary school.

Equilibrium effects. Small-scale experiments may not capture how the full population would respond to a scaled intervention. If a social norm message tells everyone they are below the norm, the norm itself shifts, which changes the message and the response.

These challenges do not undermine the RCT methodology — they are reasons to run more and better RCTs, and to approach the policy cycle with appropriate humility about the generalisability of findings.

2 2. Tax Compliance: Norms, Salience, and Simplification

2.1 2.1 The Standard Model of Tax Compliance

The standard economic model of tax compliance (Allingham and Sandmo, 1972) treats compliance as a rational decision under risk: a taxpayer files an honest return if and only if the expected penalty from non-compliance exceeds the expected gain from underreporting. This model implies a simple policy lever: increase audit rates and penalties until compliance becomes individually rational.

The empirical evidence reveals a puzzle: compliance rates are far higher than the standard model would predict, given observed audit probabilities and penalty rates. Most people pay taxes most of the time, even when the probability of audit is very low and the expected penalty from non-compliance is small. This suggests that something beyond expected-value calculation drives compliance: specifically, that many people comply because they consider it the right thing to do, or because they believe most others do so, or because the social costs of being identified as a non-complier (shame, stigma) are large.

These motivations — intrinsic moral obligation, social norm conformity, and anticipation of social judgement — are exactly the social preferences developed in Chapter 6 and the norm mechanisms documented in Chapter 9.

2.2 Social Norm Messages and Tax Compliance

The most influential field experiments on tax compliance apply social norm nudges to late filers and payers. As summarised in Chapter 9, Hallsworth et al. (2017) ran a large-scale RCT with HMRC in the United Kingdom, testing whether including social norm messages in tax letters changed payment behaviour.

Key findings:

- Letters stating "the great majority of people in the UK pay their tax on time" increased payment within 23 days by 1.3 percentage points (compared to a standard letter).
- Letters including locally-specific norms ("nine out of ten people in [your local area] pay their taxes on time") increased payment by 5.1 percentage points.
- Financially-salient messages ("most people in your situation pay immediately — those who don't may face a penalty") increased payment by 3.0 percentage points.

The experiment included over 100,000 letters and was sufficiently large to estimate effects reliably. Extrapolated to millions of late filers nationally, the revenue impact was substantial — hundreds of millions of pounds in accelerated tax receipts.

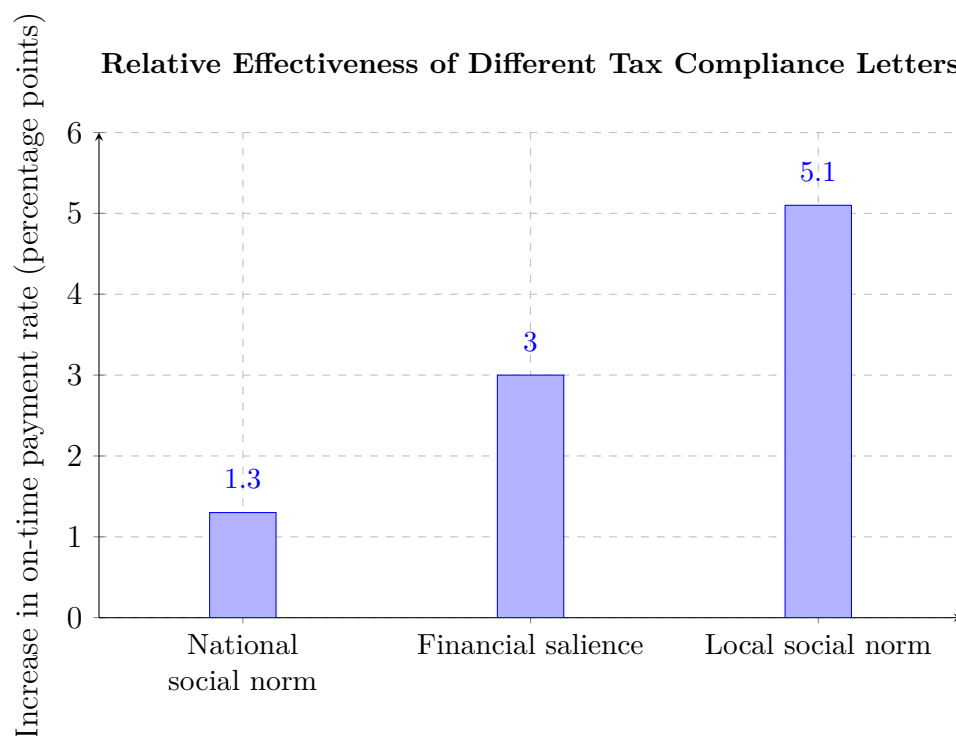
Figure 10.1: Relative Effectiveness of Different Tax Compliance Letters

Figure 1: Effect of alternative HMRC tax letters on payment behaviour. The bars show the increase in on-time payment rates relative to the standard enforcement letter. Locally targeted social norm messages generate the largest improvement, suggesting that compliance responds strongly to comparisons with similar others.

	Description
What the figure shows	A bar chart showing the increase in on-time payment rate (percentage points above the standard control letter) for three experimental letter types: national social norm (+1.3 pp), financial salience (+3.0 pp), and locally-targeted social norm (+5.1 pp).
How to interpret it	The local norm is most effective because social comparisons operate most powerfully among similar others. The pattern confirms that tax compliance is driven substantially by norm conformity — not only by expected-value calculation. Even the weakest treatment (national norm) significantly outperforms the standard enforcement letter, at near-zero additional cost.

2.3 2.3 Simplification and Tax Filing

A complementary approach targets the compliance *process* rather than the compliance *motivation*. Many taxpayers — particularly those with simple tax situations (a single employer, no investment income, no self-employment) — fail to file or file inaccurately not because they intend to evade tax but because the filing process is complex, time-consuming, and error-prone.

Pre-populated tax returns. Denmark, Norway, and Sweden pioneered the system of tax authority "pre-population": using employer-reported wage information, bank-reported interest income, and other administrative data to pre-fill tax returns. Taxpayers review the pre-filled return and either accept it or correct it.

Kleven et al. (2011) study compliance in Denmark with access to the Danish administrative data system. They find that the compliance gap — the difference between taxes owed and taxes paid — is very small for income types subject to third-party reporting (employer-reported wages, bank-reported interest) and much larger for income types that taxpayers must self-report (business income, informal income). Pre-population dramatically reduces the complexity and error rate for the majority of filers with simple situations.

Canada's CRA introduced a limited version of pre-population ("Auto-fill my return") in 2016, which allows taxpayers to automatically download information from CRA records into their tax software. Further expansion to a full pre-populated return — with opt-in confirmation rather than opt-out correction — would extend the simplification benefit to less tax-sophisticated filers.

2.4 2.4 Loss Framing and Penalty Salience

The framing of tax liabilities matters for compliance behaviour. Bhargava and Manoli (2015) study a large US government programme (the Earned Income Tax Credit) and find that changing the framing of reminder letters — from a description of benefits to a statement of the amount the household is "missing out on" — increases programme take-up substantially.

Similarly, research on penalty framing finds that emphasising the definite cost of non-compliance (present-biased individuals respond strongly to immediate penalties) rather than the probabilistic cost of audit detection (which is discounted by optimistic probability weighting) can increase compliance. This is the loss aversion principle from Chapter 3 applied to compliance: the certain loss of a penalty is more motivating than the probabilistic expected loss from audit.

3 3. Health Behaviour: From Knowledge to Action

3.1 3.1 The Knowledge-Action Gap

The domain of health behaviour presents perhaps the clearest illustration of the gap between knowing what is good for you and actually doing it. Surveys consistently reveal that the vast majority of people know they should exercise more, smoke less, drink moderately, maintain a healthy weight, and adhere to prescribed medications. The same surveys reveal that most people fall short of these goals.

Standard economics interprets this gap as reflecting time-consistent preferences: people consume cigarettes because they genuinely prefer present pleasure to future health (accounting for the future health consequences), not because they fail to follow through on their preferences. On this interpretation, policy should merely ensure that people have accurate information about health consequences; taxation and regulation are presumptuous impositions on individual choice.

Behavioural economics offers a different interpretation: the knowledge-action gap reflects present bias, inertia, the availability bias (health consequences are temporally distant and cognitively abstract), and self-control failures that people themselves recognise and regret. Most smokers say they want to quit. Most overweight people say they want to lose weight. Most physically inactive people say they intend to exercise more. The gap between intention and action is not a matter of having different preferences from what policy assumes; it is a failure of time-consistent action on acknowledged preferences.

This interpretation has very different policy implications: if the behaviour gap reflects self-control failure rather than stable preferences for the unhealthy behaviour, then interventions that help people follow through on their own stated preferences — commitment devices, default healthy options, friction reduction for healthy choices — are welfare-improving rather than paternalistic.

3.2 3.2 Smoking: The Multi-Tool Intervention

Tobacco control is the most studied domain of health behaviour policy and the most successful large-scale public health intervention in modern history. Cigarette smoking in Canada peaked at approximately 60% of adults in the 1960s and has declined to below 13% by 2023. This decline reflects the simultaneous application of multiple policy instruments:

Information disclosure. Cigarette packages in Canada have carried health warnings since 1988, and graphic photo-based warnings since 2001. Canada was among the first countries in the world to mandate graphic warnings — images of diseased lungs, oral cancer, and smoking-related consequences — covering 75% of the package.

The evidence on graphic warnings (Noar et al., 2016) finds that they are more effective than text-only warnings, that they produce emotional responses (disgust, fear) that engage System 1 processes rather than only System 2 information processing, and that they increase quit attempts among smokers. However, effects are larger for new potential smokers (reducing initiation) than for long-term addicted smokers (promoting cessation).

Plain packaging. Australia introduced standardised plain packaging in 2012, followed by the UK, Ireland, France, New Zealand, and Norway. Plain packaging removes all brand logos and colours, replacing them with standardised fonts and graphic health warnings. The Australian evidence (Cancer Council Victoria, 2016) finds a significant reduction in smoking prevalence following plain packaging introduction, though the effect is difficult to isolate from concurrent policy changes.

Taxation. Tobacco taxes are the most effective single instrument for reducing smoking prevalence. A 10% price increase reduces consumption by approximately 4–8% among adults and 8–14% among young people (who are more price-sensitive). Canadian tobacco taxes are approximately 70% of the retail price of cigarettes, contributing to the high price (approximately \$20–\$25 per pack) that substantially reduces affordability.

Default smoke-free environments. Banning smoking in workplaces, restaurants, and public places — effectively setting a smoke-free default for shared spaces — dramatically reduces secondhand smoke exposure and reduces the social reinforcement of smoking norms. The combination of the smoking ban with graphic warnings and taxes produced the steepest decline in Canadian smoking prevalence in the 2000s.

The tobacco policy toolkit illustrates a general principle: the most effective public health interventions apply multiple tools simultaneously, each targeting a different mechanism. Taxes target the rational cost-benefit calculation. Graphic warnings engage emotional System 1 processes. Plain packaging reduces the attractiveness of smoking as a social identity. Smoke-free defaults change the social environment.

3.3 Physical Activity: Commitment Devices and Temptation Bundling

Physical inactivity is among the most significant public health challenges in high-income countries. The majority of Canadians do not meet the Canadian Physical Activity Guide-

lines (150 minutes of moderate-to-vigorous activity per week for adults). Yet surveys reveal that most inactive people *intend* to be more active and *want* to exercise more. This is a textbook present bias problem: the costs of exercise are immediate (effort, discomfort, time) while the benefits are delayed (improved health, longevity).

Two novel behavioural interventions address the commitment and motivation problem:

Temptation bundling. Katherine Milkman, Julia Minson, and Kevin Volpp (2014) developed the concept of **temptation bundling**: pairing a temptation (an activity that is enjoyable but produces no long-run benefit) with a virtue (an activity that is beneficial but immediate costly). The most powerful version: individuals only access the temptation when engaging in the virtue.

Milkman et al. ran an experiment in a university gymnasium in which participants were given audio novels (the temptation) that could only be played on iPods kept at the gym. Participants who received iPods with gripping audio novels attended the gym substantially more than those who did not. The mechanism: the audio novel creates an immediate reward for gym attendance that partially offsets the immediate discomfort of exercise, helping present-biased individuals choose the virtuous action.

Commitment contracts. The Commitment to Healthy Goals (CHG) study by Volpp et al. (2008) tested whether financial commitment contracts increased weight loss among obese adults. Participants were randomly assigned to a control group (standard counselling) or one of two treatment groups: a deposit contract (participants put up \$3 per day, refunded if they met weekly weight loss goals) or a lottery contract (participants entered a daily lottery, winning \$10 with 20% probability if they met their goal). The average weight loss at 16 weeks was 3.9 pounds in the control, 13.1 pounds in the deposit contract group, and 14.0 pounds in the lottery group. The commitment contracts tripled weight loss relative to standard counselling.

3.4 Medication Adherence

Non-adherence to prescribed medication is a major health and economic problem. In Canada, approximately 50% of patients with chronic conditions do not take their medication as prescribed. Non-adherence contributes to preventable hospitalisation, disease progression, and death, at an estimated annual economic cost of over \$4 billion (Iuga and McGuire, 2014).

The mechanisms driving non-adherence overlap substantially with the mechanisms driving other present-biased behaviours: the costs of taking medication (side effects, remembering,

expense) are immediate, while the benefits (preventing future disease progression) are delayed, probabilistic, and often invisible.

Behavioural interventions targeting adherence include:

- **Simplification:** Once-daily dosing dramatically improves adherence relative to multiple-dose regimens, even when the total medication burden is identical.
- **Reminders:** Text message reminders sent at the time the patient usually takes medication improve adherence by 6–8 percentage points (Kannisto et al., 2014).
- **Financial commitment contracts:** In low-income settings, refundable deposits conditional on adherence improve adherence substantially. In high-income settings, small financial incentives for adherence (pill-bottle electronic monitors that trigger a small reward when opened) have shown effects in diabetes and HIV treatment.
- **Social accountability:** Adherence buddy programmes — in which a family member or community health worker monitors and encourages adherence — are particularly effective in high-burden disease contexts.

4 4. Consumer Finance: Beyond Disclosure

4.1 4.1 The Information Model and Its Limits

Consumer financial protection has historically relied on an information model: require firms to disclose the terms of financial products, and trust consumers to use this information to make good decisions. Truth-in-lending laws require disclosure of APR. Mortgage regulations require standardised disclosure statements. Insurance contracts must include key fact statements.

The information model rests on the standard economic model of the rational consumer: given accurate information, consumers choose the product that best serves their interests. The policy role is to ensure information accuracy and accessibility; beyond that, markets work.

Behavioural economics challenges this model on both empirical and theoretical grounds. Empirically: consumers make systematic errors in financial decisions — they do not understand compound interest, they underestimate the true cost of credit, they are

subject to framing effects in product presentation — even when information is disclosed. Theoretically: the human cognitive architecture processes information selectively, anchors on salient features (the monthly payment), discounts future consequences, and is overwhelmed by complexity.

4.2 Disclosure Regulation and Its Limits

Lauren Willis (2008, 2011) has documented the limitations of financial disclosure regulation in detail. Her central argument: disclosure regulation assumes that consumers, once informed, will use the information to make better decisions. The evidence is that this assumption is false for most consumers most of the time.

Consider mortgage disclosure. A standard Canadian mortgage disclosure statement contains the mortgage amount, the interest rate, the amortisation period, the payment frequency, the prepayment privileges, and the penalties for breaking the mortgage — across multiple pages of fine print. Research by Lacko and Pappalardo (2007) found that consumers who read mortgage disclosures still could not identify the cost of the mortgage, the penalty for early repayment, or the presence of a prepayment penalty. Disclosure was not sufficient for informed choice.

The problem is not that consumers are unintelligent. The problem is that mortgage products are genuinely complex, that the relevant comparison (total cost of credit across the life of the mortgage, adjusted for prepayment) requires calculation that most people cannot perform without computational assistance, and that the disclosed information is formatted in a way that does not highlight the features most consequential for most borrowers.

The behavioural response to disclosure failure is not to abandon disclosure but to redesign it according to behavioural principles:

- **Simplified key facts statements.** Disclosures limited to the four or five numbers that matter most for the typical consumer, presented prominently in the simplest possible terms.
- **Standardised comparison formats.** Uniform tables that allow consumers to compare across products without doing arithmetic. Mortgage calculators embedded in the disclosure document.
- **Loss framing of costs.** Presenting the cost of non-optimal choices in terms of "what you would pay more than the lowest-cost equivalent product" — a loss-framed

comparison — produces stronger response than a neutral presentation of the cost.

- **Mandatory summary comparison.** At the point of sale, requiring the retailer to provide a table comparing the offered product to the lowest-cost equivalent product available in the market.

4.3 4.3 Credit Cards: The Minimum Payment Trap

The credit card minimum payment problem — documented in Chapter 9 — illustrates how firms can use choice architecture to exploit consumer biases, and how regulation can counteract this.

Neil Stewart (2009) showed that the presence of a minimum payment figure on credit card statements anchors cardholders at a low payment level, even though paying the minimum is extremely costly over time. This is an anchoring effect (Chapter 2) with large financial consequences: a cardholder with a \$5,000 balance making only minimum payments would take over 20 years to repay the balance and would pay over \$6,000 in interest — more than the original balance.

Regulatory responses include:

- **US CARD Act (2009):** Required statements to display the time and total cost to repay the balance at the minimum payment, alongside the payment required to repay the balance in 36 months.
- **UK Financial Conduct Authority (2011):** Required statements to display a warning message and the total cost of minimum-only payments.

Evidence suggests that these disclosures reduce reliance on minimum payments among a subset of cardholders, though effects are modest and concentrated among financially-engaged consumers.

A more radical intervention: **prohibit minimum payment displays** and replace them with "suggested payment" figures that represent a more financially appropriate payment level (e.g., 1/12 of the balance, or the amount that would repay the balance in 36 months). This design uses the anchoring effect proactively rather than neutrally, directing the anchor toward a less harmful reference point.

4.4 Investment Advice and Behavioural Suitability

Chapter 8 documented that individual investors are systematically overconfident, trade too frequently, hold underdiversified portfolios, and are subject to the disposition effect — all of which reduce returns relative to passive strategies.

Financial advisors could, in principle, correct these errors. In practice, the incentive structure of the financial advice industry has historically worked against client interests: advisors who are compensated through commissions on product sales earn more by recommending high-fee actively managed products than by recommending low-fee index funds, even when the latter is better for the client.

Canada implemented client-focused reforms to the securities regulatory framework in 2021 (the "CFRs"), requiring advisors to act in the "best interest" of clients — the best interest standard — rather than merely recommending "suitable" products. This is a regulatory response to the conflict-of-interest problem in financial advice: it changes the incentive structure for advisors rather than relying on consumers to identify and correct advisors' biased recommendations.

Behavioural requirements for financial advice include:

- **Risk profiling with behavioural awareness.** Standard risk questionnaires that ask investors to state their risk tolerance produce unreliable responses because stated risk tolerance diverges dramatically from actual risk tolerance in market downturns. Behaviourally-informed risk profiling scenarios — presenting investors with realistic depictions of portfolio losses and asking how they would respond — produce more accurate risk profiles.
- **Default diversified portfolio.** For investors who do not express a specific investment mandate, the default investment option should be a well-diversified, low-fee lifecycle fund — the "best interest" default for the typical retail investor.
- **Loss aversion counselling.** Financial advisors trained to understand loss aversion can reduce clients' tendency to sell in market downturns (the typical loss-averse response) and help them maintain long-term investment strategies.

5. Environmental Behaviour: Green Defaults and Carbon Labels

5.1 The Behavioural Gap in Environmental Policy

Environmental policy faces a version of the present bias problem that is, if anything, more severe than any of the individual domains discussed so far. Climate change produces costs that are:

- **Temporally distant:** The worst consequences are projected for decades and centuries hence.
- **Geographically diffuse:** Most of the people making emissions decisions will not personally experience the worst consequences.
- **Probabilistic and uncertain:** The specific consequences are subject to scientific uncertainty.
- **Small in individual magnitude:** Any individual's emissions reduction has negligible impact on the global outcome.

Each of these features attenuates the behavioural force of pro-environmental motivation. Present bias causes people to discount future climate impacts; spatial and statistical distance reduces their emotional salience (the "identified victim effect" of Chapter 2); uncertainty triggers probability distortion; and the negligible individual impact makes free-riding individually rational even for people who genuinely care about the collective outcome.

Standard economic policy — carbon pricing — addresses the incentive problem directly: if the social cost of carbon is internalised through a carbon tax or cap-and-trade system, the private calculation of emissions decisions aligns with the social optimum. This is the correct framework.

But carbon pricing alone is insufficient for two reasons. First, it faces political constraints: carbon prices sufficient to achieve emission targets require large increases in energy costs that generate political opposition. Second, even with a carbon price, behavioural biases may prevent the full behavioral response: if people do not notice or respond to energy price signals (because of inattention, present bias in energy use decisions, or anchoring on

historical consumption patterns), the price signal will be transmitted less effectively than the rational model predicts.

Behavioural tools can complement carbon pricing by directly addressing the psychological barriers to pro-environmental behaviour.

5.2 5.2 Green Energy Defaults

As documented in Chapter 9, switching customers' default electricity tariff to 100% renewable energy — with an opt-out option for those who prefer standard (cheaper) electricity — raised renewable energy subscription rates from approximately 3% to over 80% in European utilities. This is a pure default effect: the same choice, the same price, the same options — but arranged so that the prosocial choice requires no action.

In Canada, where many utilities are publicly owned and electricity generation already includes significant hydroelectric and nuclear generation (low-carbon), the most relevant default is carbon offset purchasing, green electricity certificates, or switching to time-of-use pricing that shifts consumption away from peak hours (when marginal generation is most carbon-intensive).

A natural field experiment by Sunstein and Reisch (2014) estimated that green defaults in electricity purchasing, travel booking, and food purchasing, if applied to European consumers, would produce carbon emission reductions roughly equivalent to removing tens of millions of cars from the road — without any change in prices or mandatory requirements.

5.3 5.3 Carbon Labelling and Nutritional Equivalents

Carbon labels — standardised indicators of the carbon footprint of consumer products, placed prominently on packaging — apply the salience principle to carbon emissions. By making the carbon consequence of purchase decisions visible at the point of choice, carbon labels aim to integrate environmental costs into the System 1 evaluation of products.

The evidence on carbon labels is promising but modest. A study by Stylos et al. (2021) found that consumers who were shown carbon labels alongside standard price and nutrition information reduced the carbon footprint of their food purchases by approximately 5–7%. Effects were larger for consumers who expressed strong environmental values and for products where the carbon footprint varied substantially across comparable alternatives.

The design of effective carbon labels raises several behavioural considerations:

- **Reference point.** What is the relevant comparison? Absolute carbon quantities (kg CO₂e) are difficult for most consumers to interpret. Traffic-light systems (green/amber/red) or relative comparisons ("this product has $X\%$ lower emissions than the average product in this category") are more interpretable.
- **Salience.** Front-of-pack placement increases response relative to back-of-pack.
- **Simplicity.** Single summary indicators outperform detailed decompositions for changing purchasing behaviour.

Canada's environment-related information requirements have been weaker than those in European countries. The recent introduction of mandatory front-of-pack nutrition labelling (2026) provides a precedent for front-of-pack environmental labelling.

5.4 Messenger Effects in Environmental Campaigns

The effectiveness of environmental messaging depends significantly on who delivers the message. This is the **messenger effect**: the same information, delivered by different sources, produces different behavioural responses.

Research on energy conservation messaging finds that:

- **Credibility.** Messages from scientific organisations are more persuasive for audiences with high science trust; messages from community leaders are more persuasive for audiences with lower institutional trust.
- **Similarity.** Messages from "people like you" are more effective than messages from distant authorities. This is an application of the social comparison principle: we are most influenced by the behaviour of similar others.
- **Identity fit.** Messages framed in terms consistent with the audience's existing identity are more persuasive. Conservative audiences respond better to messages framing energy conservation as national security and energy independence; progressive audiences respond better to climate justice framing.

These findings have direct implications for environmental communication strategy: one-size-fits-all environmental messaging is less effective than targeted messaging tailored to audience values and identity.

6 6. Development Economics: Commitment Savings and Cash Transfers

6.1 6.1 Behavioural Economics and Global Poverty

Development economics has been transformed by the combination of randomised controlled trials and behavioural insights. The pioneering work of Banerjee, Duflo, and their collaborators at the Abdul Latif Jameel Poverty Action Lab (J-PAL) established the RCT as the primary tool for evaluating development interventions, while simultaneously documenting the ways in which cognitive and social factors shape economic behaviour in low-income contexts.

A key insight of behavioural development economics: the cognitive tax of poverty is real. Sendhil Mullainathan and Eldar Shafir's book *Scarcity* (2013) argues that living under conditions of resource scarcity — whether of money, time, or calories — consumes cognitive bandwidth in ways that impair decision-making across all domains. The poor are not poor because they make bad decisions; they sometimes make suboptimal decisions *because* managing scarcity consumes cognitive resources that would otherwise be available for careful deliberation.

This insight has important policy implications: reducing the administrative burden of assistance programmes, simplifying eligibility verification, and providing cognitive relief (through reminders, planning assistance, and simplified decisions) can improve both take-up of and outcomes from development programmes.

6.2 6.2 Commitment Savings in Developing Countries

The SEED experiment (Ashraf, Karlan, and Yin, 2006, described in Chapter 9) demonstrated that willingness to accept restricted access to savings — a commitment device — substantially increased savings among time-inconsistent individuals in the Philippines. Subsequent work has replicated and extended this finding across multiple developing country contexts.

Kenya: Dupas and Robinson (2013) provided small savings devices (lockboxes) to informal market vendors in Kenya, with no interest return. Simply having a dedicated place to store savings — separate from the household's general cash — increased health savings by 66% among women and 98% among market vendors. The device worked by creating

a mental account (Chapter 3) that reduced the temptation to spend earmarked health savings on non-health goods.

Malawi: Brune et al. (2016) offered commitment savings accounts to smallholder farmers in Malawi. Farmers with commitment accounts saved 53% more than control group farmers by the time the next harvest season arrived. The commitment mechanism — restricting access until the planting season — helped farmers resist within-family pressure to share cash savings with relatives, a significant barrier to savings accumulation in many developing country contexts.

Ghana: Karlan et al. (2016) tested commitment savings and goal-setting reminders among microentrepreneurs in Ghana. Entrepreneurs who received commitments *and* reminders invested 17% more in their businesses. Neither intervention alone produced significant effects; the combination was necessary.

These experiments reveal an important complementarity: commitment devices are most effective when combined with reminders (to make the commitment salient at decision moments) and with appropriate account design (to reduce the social pressures that can overwhelm individual commitment).

6.3 6.3 Conditional Cash Transfers

Conditional Cash Transfers (CCTs) are cash payments to poor households conditional on complying with requirements related to health, education, or nutrition — typically requiring children to attend school and health clinic visits. CCTs have been implemented at large scale in Mexico (PROGRESA/Oportunidades), Brazil (Bolsa Família), Colombia (Familias en Acción), and over 60 other countries.

CCTs are not strictly a nudge instrument — the conditionality requirement is a behavioural mandate (comply or lose the transfer). But their design incorporates important behavioural insights:

Present bias and undershooting: Households that value their children's education but face severe present-bias in allocating resources may underinvest in education relative to their own long-run preferences. The CCT provides an immediate financial reward for investment in the future (education), converting a delayed benefit into a current cash transfer and thereby compensating for present bias.

Information and reference effects: In contexts where the private return to education is uncertain or underappreciated, CCTs may work partly by signalling that governments

believe education is valuable — an implicit information provision alongside the direct subsidy.

Credit constraints: Where liquidity constraints prevent investment in children's health and education (a standard economic failure, not a behavioural one), CCTs address the constraint directly. Behavioural barriers and standard market failures often co-exist; the most effective interventions may target both.

The evidence on CCTs is generally very positive: Bolsa Família in Brazil, the largest CCT programme in the world, is associated with substantial reductions in poverty, improvements in school attendance, and reduced child mortality. Impact evaluations using RCT or quasi-experimental methods consistently find significant positive effects on the targeted outcomes.

7 7. Education: Information, Defaults, and Financial Literacy

7.1 7.1 Information Provision and School Choice

Education is a domain where the behavioural insight that "more information is necessary but not sufficient" is particularly clear. Many education systems offer families substantial choice among schools — charter schools, specialised programmes, different curriculum tracks — but families, particularly those with lower incomes and education, often make school choices that do not reflect their own stated preferences or the quality information available to them.

Research Study:

- **Researchers:** Christopher Hastings, Brigitte Madrian, and William Skimmyhorn
- **Year:** 2012
- **Research question:** Does providing simplified, personalised information about school quality change the schools families apply to?
- **Method:** A randomised field experiment in Charlotte-Mecklenburg, North Carolina, provided some families with simplified "report cards" summarising their assigned school's test scores and comparison to other schools in their choice set. Other families received no additional information.

- **Results:** Families in the treatment group were more likely to apply to higher-performing schools, and the improvement was concentrated among families whose current assigned school was of lower quality. Effects were larger for lower-income families and for families whose distance to higher-quality schools was shorter.
- **Economic interpretation:** Information is necessary but not sufficient: even when information is available in principle, simplifying it to a format families can easily process substantially changes behaviour. The concentration of effects among families with less access to informal information networks suggests that information interventions can reduce socioeconomic inequality in school choice.

7.2 Financial Literacy and Its Limits

Financial literacy — knowledge of basic financial concepts such as compound interest, inflation, and diversification — is strongly associated with better financial outcomes: higher savings rates, better debt management, and greater retirement wealth. It is tempting to conclude that financial literacy education is the primary policy tool for improving financial decision-making.

The evidence is more ambiguous. A comprehensive meta-analysis by Fernandes, Lynch, and Netemeyer (2014) analysed 168 papers studying financial literacy and financial outcomes. Their findings: financial literacy education explains less than 0.1% of the variance in financial behaviours, and the effects of financial literacy interventions on behaviour decay rapidly over time — the "half-life" of financial literacy training is estimated at a few months to two years.

Why are financial literacy effects so small and transient?

Just-in-time problem. Financial decisions are infrequent but consequential. Financial literacy training delivered months or years before a decision is largely forgotten by the time the decision arrives. What is needed is just-in-time information and decision support at the moment of choice.

Complexity outpaces education. Financial products are complex. Even highly financially-literate individuals make errors in evaluating complex mortgage products, annuity choices, and retirement income strategies. The solution is simplification of products, not just education of consumers.

Bounded rationality. Even educated decision-makers facing complex decisions under time pressure and cognitive load make errors. Financial literacy education helps at the

margin but cannot substitute for structural improvements in the decision environment.

The policy implication: financial literacy education is complementary to, but not a substitute for, structural interventions — default rules, simplified disclosure, mandatory cooling-off periods, fee transparency, and low-cost default investment options.

7.3 7.3 Student Loan Debt and Present Bias

Student loan debt is among the most consequential financial decisions made by young people. In Canada, approximately 50% of post-secondary students borrow to finance their education, with average debt at graduation of approximately \$17,000 (federal loans only; total debt including provincial and private loans is higher).

Repayment decisions are subject to several behavioural failures:

Present bias. Repayment decisions involve an immediate cost (loan payments) and a delayed benefit (reduced debt and interest). Present-biased borrowers delay repayment enrolment, switch to income-tested repayment plans that lower monthly payments but extend the repayment period and increase total interest, and are slow to take advantage of debt forgiveness programmes for which they are eligible.

Complexity. The federal student loan repayment system involves multiple repayment plans, income-related options, forgiveness programmes, and grace periods that are difficult to navigate without expert assistance.

Anchoring and the minimum payment. Default monthly payment amounts anchor borrowers' understanding of what "normal" repayment looks like, leading to systematic underpayment (paying only the minimum or income-based minimum) even when borrowers could afford more.

Behavioural interventions include:

- **Just-in-time counselling.** Mandatory exit counselling — delivered at the moment of graduation, when repayment is imminent — improves repayment knowledge and reduces delinquency.
- **Income-tested auto-enrolment.** Automatic enrolment in income-based repayment plans (with opt-out for those who prefer to repay faster) reduces default risk for borrowers who cannot afford standard payments.
- **Simplified comparison tools.** Online calculators that present the total cost of different repayment plans — not just the monthly payment — help borrowers see the

full consequences of their choices.

8 8. A Framework for Behavioural Policy Design

8.1 8.1 Choosing Among Policy Instruments

One of the most important questions in applied behavioural economics is when to use which policy tool. The policy design space includes at least four distinct approaches:

Standard economics (market-based). Address market failures through prices (taxes and subsidies), property rights (cap-and-trade), and information (mandated disclosure). Preserves individual choice and avoids government overreach. Works best when the market failure is the primary obstacle and when consumers can process and act on information effectively.

Libertarian paternalism (nudging). Address behavioural failures through choice architecture — defaults, simplification, social norms, framing — without restricting options or changing prices. Preserves freedom, uses knowledge of human psychology to improve outcomes. Works best when inertia, present bias, or complexity are the primary barriers, and when a welfare-improving default can be identified.

Soft paternalism. Require deliberation before a decision takes effect — cooling-off periods, mandatory counselling, waiting periods — without banning any choice. Reduces impulsive decisions and allows time-inconsistent agents to reconsider. Works best when impulsive or emotionally-driven choices are the primary concern.

Hard paternalism. Restrict or mandate choices — seatbelt laws, motorcycle helmet laws, mandatory pension contributions, minimum drinking age. Does not preserve freedom but can produce large welfare improvements when the relevant choice affects primarily the chooser and when present bias or self-control failure is severe. Appropriate when the evidence shows that the choice in question is regretted ex post by the vast majority who make it, and when nudging effects are too small.

Table 10.1: Comparing Policy Instruments for Behavioural Failures

Instrument	Mechanism	Freedom	Strength of Effect	Best For
Information provision	Inform	Full	Weak alone	Informed active choosers
Nudge (default/architecture)	Choice environment	Full	Moderate (2–80% depending on context)	Inertia, present bias, complexity
Incentive/price	Change cost-benefit	Full	Moderate to strong	Price-responsive behaviour
Soft paternalism	Delay / deliberation	Mostly	Moderate	Impulsive choices
Hard paternalism	Mandate/restriction	None	Strong	Severe self-control failures

8.2 The Mandate Threshold

When should policy go beyond nudging to mandate? There is no single answer, but several criteria are relevant:

Severity of harm. The more severe and irreversible the harm from the biased choice (death, disability, severe poverty), the stronger the case for harder intervention.

Size of nudge effects. If nudging produces only a 2–5% change in the target behaviour and the policy target is a 90% reduction, nudges are insufficient. The scale of the gap between current behaviour and the policy target matters.

Regret rates. If the large majority of people who make a particular choice subsequently say they regret it — as most addicted smokers say they regret starting — the "preference" revealed by the initial choice is a poor guide to welfare.

Third-party effects. When the biased choice harms others (children of financially irresponsible parents, passengers of drunk drivers, workers exposed to secondhand smoke), the standard liberal case for freedom of choice is weakened.

Distributional considerations. When cognitive limitations are correlated with disadvantage — when the people most subject to biased decision-making are also the poorest, least educated, and least powerful — the libertarian argument for preserving choice is weaker.

8.3 8.3 When Nudges Backfire

Not all nudge interventions succeed, and some produce counterproductive results. Understanding when and why nudges backfire is important for responsible policy design.

The boomerang effect. As discussed in Chapter 9, social norm nudges can increase the behaviour of below-norm individuals. When the norm is a bad behaviour (high consumption, high emissions, non-compliance), descriptive norm messaging can normalise the bad behaviour.

Reactance. People who feel that their autonomy is being manipulated may react against the nudge by doing the opposite. Research finds that messages framed as "telling you what to do" produce reactance (refusal) among individuals with high need for autonomy, while messages framed as "giving you information to decide" are less likely to trigger reactance.

Crowding out intrinsic motivation. Adding external incentives — even small ones — to intrinsically-motivated prosocial behaviour can reduce intrinsic motivation and net prosocial behaviour. Gneezy and Rustichini (2000) documented this in a field experiment: adding a fine for late pick-up of children at Israeli day-care centres *increased* late pick-ups, apparently because the fine converted a social obligation (moral duty not to burden the staff) into a market transaction (I can be late if I pay).

Complexity backfire. Interventions that attempt to simplify decisions by providing structured choice aids can backfire if the aid is poorly designed, if it adds complexity rather than reducing it, or if it changes the framing in ways that trigger anchoring on a wrong reference point.

9 9. The Limits of Behavioural Economics

9.1 9.1 Replication and the Credibility of Findings

Behavioural economics has been affected by the broader "replication crisis" in social science — the discovery that many published findings in psychology and economics fail to replicate when independent researchers attempt to reproduce them using similar methods.

Several high-profile behavioural economics findings have failed to replicate or have replicated with much smaller effect sizes:

- **Ego depletion** (the idea that self-control is a limited resource that depletes with use)

has not replicated reliably in large-scale pre-registered studies.

- **Power posing** (the claim that body posture affects hormones and risk-taking behaviour) has failed to replicate.
- **Some "priming" effects** (the claim that brief, unconscious exposure to a concept influences subsequent decisions) have proven unreliable.
- **Brian Wansink's cafeteria research** (on food placement and consumption) was retracted due to data problems, undermining a significant body of applied nudge research.

These failures should prompt appropriate humility about the reliability of behavioural findings — particularly those derived from small samples, non-representative populations, and non-pre-registered designs. The appropriate response is not to abandon behavioural economics but to apply higher methodological standards: pre-registration, larger samples, more diverse populations, and emphasis on replication before policy implementation.

Importantly, the most central behavioural economics findings — prospect theory, hyperbolic discounting, anchoring and adjustment, the default effect, the social norm effect — have been replicated extensively across different methodologies, populations, and contexts, and are robust.

9.2 9.2 Cultural and Contextual Limits

Behavioural economics findings vary across cultural contexts in ways that matter for policy. The ultimatum game results that show high rejection rates of unfair offers in North American and European samples are substantially weaker in other contexts: the Machiguenga of the Peruvian Amazon offer far less than half and their offers are rarely rejected (Henrich et al., 2001). Social preferences appear to vary substantially across cultural settings.

Social norm nudges that are effective in high-trust, high-compliance societies may be counterproductive in low-trust societies. If most people believe that most others *do not* comply with norms — a reasonable belief in contexts of weak institutions and low enforcement — then a social norm message stating that "most people comply" may be disbelieved and may even signal that non-compliance is normal.

Commitment savings products that work in the Philippines may not work in contexts where family networks exert irresistible social pressure to share resources. Pension defaults

that increase savings in the United States may have different effects in countries where the state pension is sufficiently generous that private supplementation is less important.

These contextual limits do not undermine the behavioural approach, but they do require:

- Field testing of interventions in the specific context of intended implementation.
- Attention to cultural mediators of behavioural effects.
- Humility about generalising from one cultural setting to another.

9.3 The Role of Institutions

Perhaps the most fundamental limitation of behavioural policy is that it focuses on individual-level cognitive and motivational failures while taking the institutional environment as given. But many of the most important economic outcomes — savings, health, education, environmental quality — are determined primarily by institutional structures: the design of pension systems, the accessibility of health care, the organisation of education, the strength of environmental regulation.

A nudge approach to retirement savings helps individuals navigate a given pension system more effectively, but it does not address the inadequacy of a pension system that provides insufficient coverage at acceptable contribution rates. A nudge approach to healthy eating helps individuals make better choices in a given food environment, but it does not address the systemic incentives that make processed food cheap, available, and aggressively marketed compared to fresh produce.

The political scientist Jacob Hacker has argued that the emphasis on behavioural nudges can serve as a politically convenient substitute for more difficult structural reforms. If people are not saving enough for retirement, the full policy response includes both automatic enrolment (a nudge) and adequate coverage levels in the public pension system (an institutional reform). Nudges are more politically tractable than institutional reform but are not a substitute for it.

Behavioural economists are generally aware of this limitation. Thaler and Sunstein's framework includes both choice architecture (nudges) and hard interventions where nudges are insufficient. But the popular discourse around nudging sometimes implies that behavioural tools are adequate substitutes for structural policy. This implication is false.

10 10. Open Questions and the Frontier of the Field

10.1 10.1 Behavioural Economics and Artificial Intelligence

The interaction between behavioural economics and artificial intelligence is one of the most rapidly evolving areas in the field. AI systems now mediate an enormous fraction of consequential decisions: social media platforms determine what information people are exposed to; recommendation systems determine what products, content, and services are suggested; algorithmic credit scoring determines who can borrow and at what rates; automated trading systems determine asset prices.

These AI systems are, in effect, choice architects at unprecedented scale. The same behavioural insights that inform nudge design — attention, salience, anchoring, social proof — inform the design of recommendation algorithms. And unlike government nudge units, which operate under democratic accountability and face public scrutiny, algorithmic choice architecture operates largely opaquely and in the interests of the platforms, not the users.

Research questions at this frontier include:

- How do recommendation algorithms shape beliefs, preferences, and decisions? (The "filter bubble" and radicalisation literature.)
- Can AI systems be designed to serve users' long-run interests rather than platforms' engagement metrics?
- How should algorithmic choice architecture be regulated? Can disclosure requirements, algorithmic auditing, or user controls improve outcomes?

10.2 10.2 The Neuroscience of Decision-Making

The dual-process model (System 1 / System 2) introduced in Chapter 1 is a theoretical abstraction. Neuroscience is beginning to provide a more mechanistic account of the neural processes underlying economic decisions — how the brain evaluates options, discounts future consequences, processes risk, and integrates social information.

Research using functional MRI, lesion studies, and pharmacological interventions has identified brain regions associated with:

- **Reward evaluation:** The ventromedial prefrontal cortex (vmPFC) and ventral stri-

tum appear to compute the subjective value of options.

- **Self-control:** The dorsolateral prefrontal cortex (dlPFC) is associated with inhibitory control over impulsive responses.
- **Intertemporal choice:** Present bias appears to be associated with differential activation of limbic regions (immediate rewards) versus prefrontal regions (future consequences).
- **Social valuation:** The temporoparietal junction (TPJ) is involved in perspective-taking and computing others' welfare.

These neural findings do not yet translate directly into policy — it is not yet feasible to tailor nudges to individuals based on their neural characteristics — but they provide a mechanistic foundation for predicting which individuals are most subject to which biases, and which interventions are most likely to engage the relevant neural processes.

10.3 10.3 Heterogeneity and Personalised Nudges

One of the most important methodological advances in applied behavioural economics is the ability to detect and respond to heterogeneity in treatment effects: different people respond differently to the same intervention, and understanding who responds to what is essential for targeting interventions effectively.

Machine learning methods applied to large administrative datasets can identify the individuals most likely to respond to a particular nudge, allowing:

- **Personalised default assignment.** Rather than one default for everyone, the default can be set based on predictable variation in preferences (different investment defaults for workers with different age, income, and tenure).
- **Targeted messaging.** Social norm messages can be personalised to the most relevant peer comparison group, rather than using a one-size-fits-all norm.
- **Sequencing.** Some interventions (information provision) are more effective when delivered before others (default options), and the optimal sequence may vary by individual.

This personalisation agenda raises privacy concerns — using individual data to design targeted behavioural interventions — that are not yet fully resolved, but represents the direction of the applied research frontier.

10.4 10.4 Long-Run Effects

Most behavioural economics experiments measure outcomes over short horizons — weeks, months, or at most a few years. The long-run consequences of behavioural interventions are much less studied, for the practical reason that it is difficult to maintain experimental designs over long periods.

Two important long-run questions:

Do nudge effects persist? Automatic pension enrolment increases participation rates sharply at the time of enrolment, but do enrollees maintain their savings rates over years and decades? Do they eventually reach savings rates comparable to what they would have achieved under an opt-in system, or does the initial default lock in long-run outcomes? The evidence to date (Chetty et al., 2014) suggests that automatic enrolment increases total savings permanently for the majority of enrollees, at least over the 10-year horizon studied.

Do nudges change preferences? If nudges lead people to adopt prosocial behaviours — saving more, exercising more, eating healthier — does repeated performance of these behaviours over time change underlying preferences in the same direction? There is suggestive evidence that "doing good" reinforces the identity of "being a good person," which increases intrinsic motivation for prosocial behaviour. But this remains an underresearched question.

11 11. Behavioural Policy: An Integrated Assessment

11.1 11.1 What Has Behavioural Economics Contributed to Policy?

Looking back over the development of the field — from the laboratory findings of the 1970s to the large-scale field experiments and government nudge units of the 2010s and 2020s — what is the cumulative contribution of behavioural economics to policy?

Improved retirement savings. Automatic enrolment in pension schemes is now standard in many countries. In the United Kingdom, the 2012 auto-enrolment mandate has enrolled over 10 million new pension savers. In Canada, the CPP enhancement moves in a similar direction. The accumulated evidence on default effects in retirement savings is among the most policy-relevant findings in modern economics.

Organ donation. Several countries — including the United Kingdom, Spain, Austria, France, and Belgium — have moved to opt-out organ donation defaults, dramatically

increasing effective donation rates and saving thousands of lives annually that would otherwise be lost to organ shortages.

Tax compliance. Social norm letters, simplified notices, and pre-populated returns have reduced the tax compliance gap and improved voluntary compliance in multiple countries, generating billions in additional revenue.

Energy conservation. Social norm home energy reports (the OPOWER model) have produced documented energy savings in millions of households, at a cost substantially below supply-side alternatives.

Financial regulation. Automatic enrolment in default investment funds, simplified disclosure requirements, best-interest standards for financial advisors, and the elimination of commission incentives in some markets (the UK's Retail Distribution Review) reflect behavioural insights about investor psychology and the limits of disclosure-only regulation.

Public health. Graphic cigarette warnings, plain packaging, and smoke-free defaults have contributed to dramatic reductions in smoking prevalence. Calorie labelling at restaurants is now mandatory in multiple jurisdictions, with measurable effects on consumption.

Development. Commitment savings accounts, just-in-time counselling, and simplified conditional cash transfer programmes have improved outcomes for millions of people in low-income countries.

These are real and substantial achievements. They do not represent a complete revolution in economic policy, but they represent a significant and growing contribution.

11.2 11.2 The Appropriate Scope of Behavioural Policy

The appropriate scope of behavioural policy is not unlimited. The case for behavioural interventions is strongest when:

1. The target behaviour reflects a well-documented behavioural failure (not merely a preference the policymaker considers suboptimal).
2. The intervention preserves genuine freedom of choice.
3. The intervention has been tested and shown to be effective in the relevant context.
4. The intervention is transparent, democratically accountable, and subject to revision if it produces unintended consequences.
5. The size of the behavioural intervention matches the scale of the problem.

The case is weakest when the target behaviour reflects genuine stable preferences rather than self-control failure, when the "better" outcome is contested, when the intervention exploits biases in ways that are not transparent to those being nudged, or when the scale of the problem requires structural change that nudges cannot achieve alone.

Behavioural economics is best understood as a complement to standard economics and to structural policy, not a substitute. It enriches our understanding of individual decision-making in ways that improve both economic theory and economic policy. But it does not replace the fundamental insight of economics — that incentives matter, that scarcity is real, and that institutional design determines long-run outcomes.

Critical Thinking Questions

11.3 Conceptual Questions

1. The "empirical policy cycle" describes a progression from laboratory finding to field pilot to scaled implementation to iteration. At which stage do you think the greatest risk of error occurs? What safeguards would you build into each stage?
1. The Allingham-Sandmo (1972) model of tax compliance predicts that most people should evade taxes given low audit probabilities. Yet empirical compliance rates are much higher than the model predicts. Does this suggest the model is wrong, or that the model is right but missing a variable? What variable would you add?
1. Financial literacy education has weak and transitory effects on financial behaviour (Fernandes et al., 2014). Does this mean financial literacy education is not worth pursuing? Or does it mean we should not expect education alone to be sufficient? What does it imply for the design of financial consumer protection policy?
1. The study of commitment savings (SEED, Ashraf, Karlan, and Yin, 2006) finds that the people who choose commitment accounts are those who are most time inconsistent. This is evidence of self-knowledge about present bias. If people know they are present biased, why don't they correct for it without needing external commitment mechanisms?
1. The evidence on conditional cash transfers (CCTs) is generally very positive: they increase school attendance, reduce child mortality, and reduce poverty. But some economists argue that CCTs are inferior to unconditional cash transfers (UCTs) because

they restrict poor households' choices. Evaluate this argument using both standard economics and behavioural economics frameworks.

1. What is the "mandate threshold" — the point at which policy should move from nudging to mandating? Develop a set of criteria for this threshold and apply them to three specific policy problems: (a) retirement savings, (b) motorcycle helmets, (c) vaccination.
1. The evidence on crowding out (Gneezy and Rustichini, 2000) suggests that adding small financial incentives to intrinsically-motivated prosocial behaviour can reduce net prosocial behaviour. Does this mean we should never use financial incentives for prosocial behaviour? What conditions determine when incentives help and when they crowd out intrinsic motivation?
1. Behavioural economics has been criticised for taking the institutional environment as given and focusing on individual-level interventions that cannot substitute for structural reform. Is this a fair critique? Can you identify a policy domain where nudging is adequate and one where it is clearly insufficient?
1. Pre-populated tax returns — where the government uses administrative data to fill in the return and the citizen approves — are used in Scandinavia. Some critics argue this gives government too much power and reduces citizen engagement in understanding their taxes. Evaluate this tradeoff from a behavioural economics perspective.
1. The interaction between behavioural economics and AI raises the question of whether algorithmic recommendation systems are "nudging" users or "manipulating" them. What criteria would you use to distinguish between nudges and manipulation in this context?

11.4 Application Questions

1. A Canadian province wants to reduce the rate of type 2 diabetes through improved dietary behaviour. The province is considering three interventions: (a) a sugar tax, (b) mandatory calorie labelling in all food service settings, and (c) default healthy meal options in provincial workplace cafeterias. Using the behavioural policy framework developed in this chapter, compare these interventions in terms of mechanism, likely effectiveness, and ethical concerns.
1. Design a behavioural intervention programme for increasing physical activity among middle-aged Canadian adults. Your programme should include at least three different tools (one commitment device, one social norm tool, and one simplification/default

tool). For each tool, identify the specific mechanism and predict the likely effect.

1. The Government of Canada wants to increase RRSP and TFSA contribution rates among Canadians aged 25–40. Using the EAST framework (Easy, Attractive, Social, Timely), design a programme that does not involve changing contribution limits or tax incentives. For each element of EAST, specify what would change and why it would be effective.
1. A study finds that social norm messaging in Canadian CRA letters increases on-time payment by 3 percentage points. The letter is sent to 2 million late filers. What is the total increase in on-time payments? If the average outstanding payment is \$1,200, what is the total additional revenue collected? What is the approximate cost of the intervention, and is it cost-effective compared to a standard enforcement letter?
1. You are advising a major Canadian bank on improving financial outcomes for its retail banking customers. The bank currently does not use any behavioural interventions. Identify three specific behavioural failures among retail banking customers and design an intervention for each. For each intervention, specify how the bank would know if it was working.
1. Design an RCT to test whether a commitment savings feature in a Canadian mobile banking app increases monthly savings deposits. Specify: (a) the target population, (b) the treatment and control conditions, (c) the randomisation strategy, (d) the primary outcome measure, (e) the sample size needed (you may estimate), and (f) how you would assess whether the effect persisted after 12 months.
1. A charitable organisation wants to increase the average donation from its existing donors (most of whom give \$50 annually) to \$100. Using the behavioural concepts of this chapter, design a communication strategy that uses at least three different mechanisms. Predict the likely effect size for each mechanism.
1. The Canadian federal government is designing a new student debt repayment system. Using the behavioural insights of this textbook, identify the three most important behavioural barriers to optimal repayment and design a choice architecture that addresses each. How would the new system differ from the current system?
1. A public health authority wants to reduce opioid overdose deaths in a Canadian city. One option is a social norm campaign ("most people who use opioids do not mix them with alcohol"). Another is a commitment device (asking users to carry naloxone and

call the help line if they feel the urge to use alone). Evaluate both options in terms of behavioural mechanism, likely effectiveness, and ethical considerations.

1. Canada's climate targets require a 40–45% reduction in greenhouse gas emissions by 2030 relative to 2005 levels. Using the distinction between nudges, incentives, and mandates developed in this chapter, design a policy package that uses each type of instrument for a specific aspect of emission reduction. Which instrument do you think will be most important to achieving the target, and why?

11.5 Discussion Questions

1. Sendhil Mullainathan and Eldar Shafir argue in *Scarcity* that poverty imposes a "cognitive tax" that impairs decision-making. If true, what are the implications for evaluating the choices of poor people? Does it change your assessment of policies that treat poor people's choices as reflecting stable preferences?
1. Behavioural economics began as a critique of standard economics. Is it now becoming the new mainstream? What has been gained and what might have been lost in the transition from a heterodox critique to an establishment policy tool?
1. The UK Behavioural Insights Team was initially entirely government-owned and has evolved into a social purpose company that advises both governments and private clients. Some critics argue that this commercialisation creates conflicts of interest. Others argue that it improves efficiency. Evaluate both views.
1. Development economists have used RCTs to evaluate specific interventions (deworming, bed nets, commitment savings) in ways that have had significant policy influence. But some critics argue that this focus on what works at the micro-level has distracted from understanding *why* some countries are rich and others are poor — a question that cannot be answered with RCTs. Evaluate this critique.
1. The 2019 Nobel Prize in Economics went to Banerjee, Duflo, and Kremer for "their experimental approach to alleviating global poverty." The 2017 Nobel Prize went to Thaler for behavioural economics. Do these two research programmes complement each other, compete with each other, or address fundamentally different questions?
1. A government nudge unit finds that a social norm message increases tax compliance by 5 percentage points. A more intensive enforcement programme (additional audits, higher penalties) increases compliance by 15 percentage points but costs 50 times as

much per additional dollar collected. Should policy prioritise the cheap nudge or the costly enforcement? What additional information would you need?

1. "Behavioural economics has shown that people are predictably irrational. But it has not shown that governments or corporations are any more rational. Why should we trust irrational governments to design better choice architectures for irrational individuals?" Evaluate this argument. Is it a sufficient objection to the libertarian paternalist project?
1. Should behavioural economics be taught in secondary school? What would be the benefits and risks of widespread public literacy about cognitive biases, heuristics, and nudges? Could it undermine the effectiveness of nudges if people know they are being nudged?
1. The frontier of behavioural economics is increasingly quantitative — machine learning, large administrative datasets, causal identification through natural experiments. Is this a healthy direction for the field, or does it risk losing the psychological depth that made early behavioural economics distinctive?
1. Reflecting on all ten chapters of this textbook, identify the single insight from behavioural economics that you believe has the greatest potential to improve human welfare if applied at scale. Justify your choice with reference to the evidence reviewed in the course.

Review Questions

Definitions:

1. What is a randomised controlled trial (RCT)? What is the fundamental advantage of random assignment over observational studies?
2. Define the "behavioural policy cycle" and describe its six stages.
3. What is the "knowledge-action gap" in health behaviour, and how does behavioural economics explain it differently from standard economics?
4. Define "temptation bundling" and explain the psychological mechanism behind it.

5. What is the "mandate threshold" in behavioural policy design? What criteria determine when policy should move from nudging to mandating?

Short Answers:

1. Describe the Hallsworth et al. (2017) tax compliance experiment. What was the treatment, what was the control, and what were the main findings?
2. Explain why financial literacy education has weak effects on financial behaviour (Fernandes et al., 2014). What does this imply for consumer financial protection policy?
3. Describe the SEED commitment savings experiment (Ashraf, Karlan, and Yin, 2006). What was the intervention, who benefited most, and what does the result tell us about self-knowledge of present bias?
4. What is the "cognitive tax" of poverty (Mullainathan and Shafir, 2013)? What does it imply for evaluating the economic choices of poor people?
5. Identify three specific conditions under which nudges backfire. For each, describe the mechanism and identify a corrective.

Analytical Questions:

1. A government is considering three approaches to reducing sugar consumption: (a) a sugar tax, (b) a social norm campaign, (c) a default healthy meal option in government canteens. For each approach, identify the primary behavioural mechanism, predict the likely effect size (based on the evidence in this chapter), and identify a potential unintended consequence.
2. Explain the difference between hard paternalism and libertarian paternalism using two examples from this chapter. When does the libertarian paternalist response suggest that hard paternalism is appropriate?

Applied Scenarios:

1. You are a policy advisor to the Minister of Finance of Canada. The Minister wants to increase the proportion of Canadians saving adequately for retirement. Design a three-part behavioural policy package that uses: (a) a default intervention, (b) a simplification intervention, and (c) a social norm intervention. For each, explain the specific mechanism, the evidence base, and how you would evaluate its effectiveness.

2. A hospital wants to reduce missed medical appointments ("no-shows") among outpatients. Currently, 25% of appointments are missed. Using the EAST framework, design a behavioural intervention to reduce the no-show rate to below 10%. Predict the likely effect of each element of your design.

Chapter Summary

This final chapter has developed a comprehensive framework for translating behavioural economics insights into policy and practice.

The empirical policy cycle. Sound behavioural policy moves from laboratory finding to theoretical mechanism to field pilot to replication to scale-up to iteration. The randomised controlled trial is the methodological cornerstone: it provides causal identification by randomly assigning subjects to treatment and control conditions. Without rigorous field testing, laboratory findings cannot be assumed to generalise to policy contexts.

Tax compliance. The standard model predicts much lower compliance than we observe; intrinsic motivation, social norms, and anticipated social costs drive the gap. Social norm letters (Hallsworth et al., 2017) increase on-time payment by 1.3–5.1 percentage points, with locally-targeted norms most effective. Pre-populated tax returns reduce complexity and error rates dramatically. Loss framing of penalties increases salience of compliance cost.

Health behaviour. The knowledge-action gap — knowing what is healthy but failing to act accordingly — reflects present bias, inertia, and the abstractness of future health consequences. Effective interventions include graphic warnings (engaging System 1 rather than System 2), smoke-free defaults (normalising prosocial behaviour), temptation bundling (Milkman et al., 2014), and commitment contracts (Volpp et al., 2008, which tripled weight loss relative to standard counselling).

Consumer finance. Disclosure regulation is necessary but insufficient: research consistently finds that consumers cannot effectively use disclosed information in complex financial decisions. Behavioural alternatives include simplified key facts statements, default diversified investment funds, cooling-off periods, and best-interest standards for financial advisors.

Environmental behaviour. Carbon pricing is the correct structural response to climate

change, but behavioural tools complement it by addressing the attention and salience barriers to pro-environmental behaviour. Green energy defaults raised renewable subscription rates from 3% to over 80%. Carbon labels reduce the footprint of food purchases by 5–7%. Social norm messages (Allcott, 2011) reduce energy consumption by approximately 2% at a cost substantially below supply-side alternatives.

Development economics. Commitment savings accounts (Ashraf, Karlan, and Yin, 2006; Dupas and Robinson, 2013) increase savings by 50–98% among time-inconsistent individuals in low-income countries. Simplified conditional cash transfers address the interaction of present bias with credit constraints and improve school attendance, child health, and long-run income. The cognitive tax of poverty means that reducing administrative burden — simplification — can dramatically increase take-up of existing programmes.

Education. Personalised, simplified information about school quality changes school application behaviour, with the largest effects for the most disadvantaged families. Financial literacy education has weak and transitory effects; simplification of financial products and decisions is more effective than education alone.

Policy instrument choice. The policy design space includes standard economics (information and prices), nudges (choice architecture), soft paternalism (deliberation requirements), and hard paternalism (mandates and restrictions). The mandate threshold — when to move from nudging to mandating — depends on severity of harm, size of nudge effects, regret rates, third-party effects, and distributional considerations.

Limits of behavioural economics. The replication crisis has affected some behavioural findings; the most robust findings (prospect theory, hyperbolic discounting, default effects, social norms) have survived extensive replication. Cultural context matters: interventions must be tested in the specific context of intended implementation. Behavioural economics complements but cannot substitute for institutional reform when structural factors are the primary determinant of outcomes.

The frontier. Key open questions include: how AI recommendation systems interact with and exploit cognitive biases; whether personalised nudges can be designed to serve individual interests rather than platform interests; whether nudge effects persist over long time horizons; and how behavioural insights should inform the regulation of algorithmic choice architecture.

Glossary

Behavioural Policy Cycle. The iterative process by which laboratory findings in behavioural economics are translated into government policy: (1) laboratory finding, (2) theoretical mechanism, (3) field pilot, (4) replication and variation, (5) scale-up and evaluation, (6) iteration. Each stage requires specific methodological tools and carries specific risks of error.

Cognitive Tax of Poverty. The argument (Mullainathan and Shafir, 2013) that managing resource scarcity consumes cognitive bandwidth, leaving less capacity for careful deliberation across all decision domains. Implies that the poor are not poor because they make bad decisions but that managing poverty itself impairs decision-making — with implications for programme design (reduce administrative burden) and welfare assessment (contextualise apparently poor decisions).

Conditional Cash Transfer (CCT). A social programme that provides cash payments to low-income households conditional on specified behaviours — typically school attendance and health clinic visits. Addresses the interaction of present bias with liquidity constraints: the cash transfer provides an immediate reward for investing in future-oriented activities. Largest example: Brazil’s Bolsa Família, reaching 14 million families.

Crowding Out of Intrinsic Motivation. The phenomenon whereby adding external incentives to intrinsically-motivated prosocial behaviour reduces net prosocial behaviour by converting a moral obligation into a market transaction. Documented by Gneezy and Rustichini (2000) in the daycare fine experiment: adding a fine for late pickup increased late pickups by transforming a social norm into a price signal.

Hard Paternalism. Policy that restricts or mandates choices — banning options or requiring specific behaviours — regardless of the individual’s preference. Examples: seatbelt laws, motorcycle helmet laws, mandatory pension contributions, minimum drinking age. Justified when the behaviour in question causes severe and often-regretted harm primarily to the decision-maker, and when softer interventions have insufficient scale.

Just-in-Time Information. Information provided at the moment it is decision-relevant, rather than in advance. More effective than advance information because people forget information between learning and decision, and because attention is focused on the relevant domain at the moment of decision. Examples: point-of-purchase calorie labels, mortgage comparison tools at the point of signing, exit counselling for student debtors.

Knowledge-Action Gap. The empirical regularity that people frequently know what health behaviour is optimal but fail to act accordingly. Explained by behavioural economics as reflecting present bias (immediate costs of healthy behaviour outweigh immediate benefits), inertia, and the abstractness of future health consequences. Challenges the standard economic interpretation of health behaviour choices as reflecting stable preferences.

Mandate Threshold. The point at which behavioural policy should move from nudging to mandating — restricting or requiring choices rather than merely making the preferred choice easier. Depends on severity of harm, size of achievable nudge effects, regret rates, third-party effects, and distributional concerns. Not a fixed rule but a judgement that must weigh these factors case by case.

Messenger Effect. The finding that the effectiveness of a message depends substantially on who delivers it, not just what it says. More credible, similar, and identity-compatible messengers produce stronger behavioural responses. Implication for environmental campaigns: message tailoring to audience values and identity is as important as message content.

Pre-populated Tax Returns. The system, used in Scandinavian countries, in which the tax authority uses third-party-reported income data (employer wages, bank interest) to fill in the tax return; citizens review and either accept or correct the return. Reduces complexity, errors, and filing burden. Evidence (Kleven et al., 2011) shows that compliance is highest for income types subject to third-party reporting.

Publication Bias. The tendency of academic journals to publish positive findings (showing that an intervention works) more readily than null results (showing no effect). Means the published literature overestimates the average effectiveness of interventions. Meta-analyses of behavioural interventions must account for publication bias, typically by weighting larger, pre-registered studies more heavily.

Randomised Controlled Trial (RCT). The methodological gold standard for causal inference: subjects are randomly assigned to treatment (receives the intervention) and control (does not) groups, and outcomes are compared. Random assignment ensures that treatment and control groups are identical in expectation on all characteristics, so that observed differences in outcomes can be attributed to the intervention. Now the dominant method for evaluating behavioural policy interventions.

Reactance. The psychological phenomenon whereby individuals who feel their autonomy is being undermined by a directive or paternalistic intervention respond by doing the opposite of what is requested. A risk of explicitly directive nudges; reduces by framing

interventions as information provision rather than behavioural direction.

Replication Crisis. The finding, beginning around 2011, that many published findings in psychology and economics fail to replicate when independent researchers attempt to reproduce them. Has affected several high-profile behavioural economics findings. The appropriate response: higher methodological standards (pre-registration, larger samples, replication requirements), not abandonment of behavioural science.

Soft Paternalism. Policy that requires deliberation before a choice takes effect — cooling-off periods, mandatory counselling, waiting periods — without restricting what can ultimately be chosen. Addresses impulsive choices and present bias without removing freedom. Example: 10-day cooling-off period for financial product purchases.

Temptation Bundling. The strategy of pairing a temptation (immediately enjoyable but long-run neutral or harmful activity) with a virtue (immediately costly but long-run beneficial activity). Accessing the temptation is conditional on engaging in the virtue. Developed by Milkman, Minson, and Volpp (2014): audio novels accessible only at the gym increased gym attendance. Mechanism: immediate reward for the virtuous action partially compensates for its immediate cost.

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End of Chapter 10

End of Textbook. You have now completed *ECON 230: Introduction to Behavioural Economics*. From the rational choice model and its empirical failures (Chapter 1), through heuristics (Chapter 2), prospect theory (Chapter 3), time inconsistency (Chapter 4), savings behaviour (Chapter 5), social preferences (Chapter 6), behavioural game theory (Chapter 7), behavioural finance (Chapter 8), nudges (Chapter 9), and behavioural policy design (Chapter 10), you have built a comprehensive understanding of how people actually make decisions and what this means for economics, policy, and everyday life. The central insight — that humans are neither perfectly rational nor perfectly irrational, but systematically, predictably imperfect in ways that can be understood, modelled, and in many cases improved — is one of the most practically important ideas in modern social science.